

SMART VEHICLE SAFETY

PHASE 5 — TESTING & VALIDATION REVIEW (TVR)

Smart Vehicle Accident Detection and Emergency Rescue System

Objective: prove that the prototype functions as designed through documented, repeatable and measurable testing.

1. Functional Testing Report

ID	Test condition	Expected result	Pass
T01	Power-up	ESP32 and MPU6050 initialize	■
T02	Stationary	NORMAL state	■
T03	Hard-braking-like bench disturbance	No confirmed accident	■
T04	Pothole-like simulated disturbance	No confirmed accident without confirmation	■
T05	Warning + button	CANCELLED/NORMAL	■
T06	Confirmed controlled test	ACCIDENT state	■
T07	GPS unavailable	Invalid/degraded GPS state	■
T08	Network unavailable	Local detection continues	■
T09	Network restored	Event/latest data syncs	■
T10	Dashboard listener	Realtime update visible	■
T11	Power restart	System returns to monitoring	■
T12	Repeated event	No uncontrolled duplicate alerts	■

2. Performance Evaluation

Metric	How to measure	Record
Detection latency	Threshold crossing → warning/confirmed state	Mean/max
False-positive rate	False accidents / non-crash trials	Measured %
False-negative rate	Missed positive trials / positive trials	Measured %
GPS acquisition time	Power-on → valid fix	Mean/max
Cloud update latency	Event creation → dashboard update	Mean/max
Battery endurance	Runtime under measured load	Hours
Recovery time	Network/power interruption → normal operation	Seconds

3. Test Data Sheet

Trial	Scenario	Peak G	Tilt°	GPS	State	Result
1	Stationary				NORMAL	
2	Hard braking simulation				NORMAL/WARNING	
3	Pothole simulation				NORMAL/WARNING	
4	Confirmed controlled test				ACCIDENT	
5	Warning cancelled				CANCELLED	
6	GPS unavailable			No	DEGRADED	

Trial	Scenario	Peak G	Tilt°	GPS	State	Result
7	Network unavailable				LOCAL ONLY	
8	Network restored				SYNCED	

Fill all blank cells using actual measured values. Never fabricate accuracy, latency or reliability results.

4. Error Analysis

Problem	Likely cause	Corrective action
False accident on pothole	Threshold too low	Use impact + tilt + persistence
Hard braking warning	Acceleration spike	Require confirmation before escalation
GPS no fix	Poor satellite visibility	Show invalid/degraded state
Cloud failure	Network unavailable	Retry/queue event
Sensor drift	Calibration/mounting	Calibrate and rigidly mount
Repeated alerts	No lockout/rearm	Add event lockout/acknowledgement
Random reset	Supply dip	Verify battery/buck wiring

5. Debugging Report

Bug	Symptom	Root cause	Fix	Verification
D01	MPU not detected	I2C wiring/address	Check SDA/SCL/address	Sensor test
D02	GPS no fix	Indoor/antenna conditions	Outdoor test	Valid fix
D03	Firebase permission denied	Rules/authentication	Authenticate + correct path	Read/write succeeds
D04	Dashboard stale	Wrong path/listener	Correct reference	Realtime update
D05	False alarm	Logic too sensitive	Tune with test data	Repeat test

6. Power Consumption Analysis

Mode	Voltage (V)	Current (A)	Power (W)
Idle/monitoring			$V \times I$
GPS active			$V \times I$
Wi-Fi active			$V \times I$
Buzzer active			$V \times I$
Peak event			$V \times I$

Measure each mode using the actual prototype. Runtime estimate $\approx$ usable battery capacity (Ah) $\div$ average current (A), adjusted for converter losses and safety margin.

7. Reliability Testing

Run repeated normal-operation trials, non-crash bench/simulation disturbances, repeated positive controlled tests, power-cycle tests and temporary network-loss tests. Record every trial and outcome. Do not perform deliberate road collisions or dangerous driving tests.

8. Acceptance Criteria

Area	Acceptance condition
Hardware	No intermittent/loose connections
Detection	Defined normal/warning/accident states behave consistently
Cancellation	Warning can be cancelled
GPS	Valid coordinates shown when fix exists

Area	Acceptance condition
Connectivity	Cloud sync works when network exists
Dashboard	Correct realtime state/location
Documentation	Measured evidence matches prototype

9. Future Improvements

Area	Improvement
Connectivity	Independent cellular communication for wider coverage
Offline operation	Non-volatile event buffer + retry queue
Sensing	Higher-range accelerometer if testing proves necessary
Decision quality	Add speed/vehicle-state inputs and improved sensor fusion
AI/ML	Collect labelled data, train model and validate precision/recall/F1
Security	Authentication, protected rules and secret management
Validation	Larger controlled dataset and independent test set

10. Mentor-Ready Validation Statement

The system is an engineering prototype. Thresholds must be validated experimentally, and no medical or emergency-response guarantee should be claimed. If AI/ML is introduced later, report the dataset, model, training method, independent test results and metrics rather than only stating an accuracy percentage.

11. Phase 5 Submission Checklist

Deliverable	Status
Functional Testing Report	■
Performance Evaluation	■
Test Cases	■
Error Analysis	■
Debugging Report	■
Power Consumption Analysis	■
Reliability Testing	■
Future Improvements	■